How Blockchain Extends the Life of Art

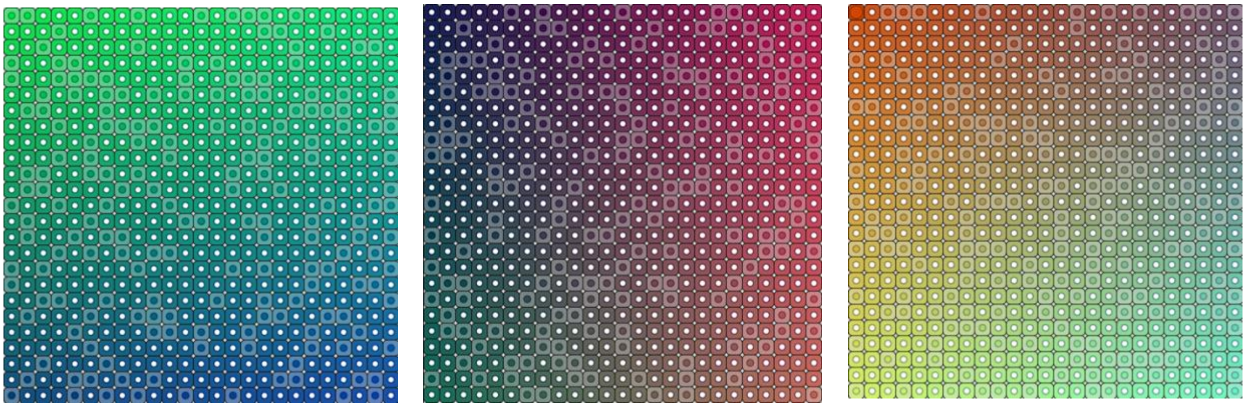

Figure 1.
https://editor.p5js.org/carbudar/sketches/ikVkKfFTn

In the contemporary art world, generative art has challenged the notions of authorship, permanence, and visual form. Rather than treating art as a static object, generative art practices often use systems, like code, rules, and randomness to produce work that unfold unpredictably. With the vast rising of blockchain technology, artists are now empowered to create artworks that continuously lives after purchase, and potentially create space for viewer-art interaction. This pushes the discussion of what it means to own art.

Generative art is commonly known by Phillip Galanter's definition, describing it as a practice where "the artist uses a system, such as a set of natural language rules, a computer program, a machine, or other procedural invention, that is set into motion with some degree of autonomy, thereby contributing to or resulting in a completed work of art."[1] With this concept, the artist gives up part of the control to the system. The blur between designing art and the

[1] Philip Galanter, "Generative Art Theory," in *A Companion to Digital Art*, ed. Christiane Paul (Hoboken, NJ: Wiley, 2016), https://ems.andrew.cmu.edu/2016-60212/resources/galanter_generative.pdf.

systemic execution allows the output to be both surprising, yet still recognizable.[2] Maintaining balance between randomness and order.

Through digital tools, generative art gained flexibility and time. Many generative arts are not shown as a finished image, rather as an unfolding performance, based on how the system is running. This time-based dynamic characteristic differentiates generative artworks to traditional art or even static digital art, where the meaning and form has no ability to become continuous.[3]

Dot Map is a long-form generative art that uses a Javascript algorithm to create a 25 x 25 matrix that puts colors in random circles within the squares. This art explores how pseudo-randomness, structure, and computation collaborate to create uniqueness. Using the modulo operator, each circle is assigned with a random number. The system decides which circles in the grid becomes colored, and which ones stays white. This rule creates a pattern that feels random while still following a consistent logic.

Every generated Dot Map becomes a snapshot of computational chance. Instead of a fixed output, Dot Map embraces the variation within the pattern. With the pre-determined parameters, this artwork allows order and unpredictability to coexist, inviting viewers to consider the variety of output that a system can produce.

Beyond its visual output, Dot Map also creates a sense of time. Viewers wait as the colors reveals itself square by square, watching the system produce the art instead of receiving a finished result. Once the pattern fully appears, the piece continues to function as an interactive

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[2] Philip Galanter, "What Is Generative Art? Complexity Theory as a Context for Art Theory," in *Generative Art Conference Proceedings* (Milan: Politecnico di Milano, 2003), https://www.generativeart.com/on/cic/papersGA2003/a22.pdf.

[3] Julia Kaganskiy, "Decoding Generative Art," *Infinite Images* (Toledo Museum of Art), July 2025, https://infiniteimages.toledomuseum.org/essay/decoding-generative-art.

environment rather than a static image. Users can still engage with the system by hovering their mouse over each grid to enlarge the circles. Dot Map does not end when the randomness stops, it remains alive, inviting interaction that can only happen through the blockchain medium.

When a generative art leverages the blockchain, and stores its source code on the chain, the art can continue to live after the ownership. Each minting process results in a different output for different owners. This mixture of code and blockchain has the ability to “breathe a new life” into the medium and bring it “to a wider audience”.[4] The static ownership of a fixed output is replaced dynamic ownership of a system. This allows the system to become the art and to become its own living ecosystem that has the ability to create infinite amounts of art which are traceable and permanent.

This shift transforms the nature of art ownership. Traditional art’s value and meaning are often tied to a moment, historic, or monetary value. On the other hand, generative blockchain-based art constantly renews itself. Every execution of the code, and each interaction after ownership creates a new moment and prolongs the life of the art. Ownership of the art becomes a participatory experience. The owner is not just a collector, but also takes part in the creation and conservation of the blockchain-based generative art. Specifically with Dot Map, the hover interaction stays animated even after the system stops running, increasing the arts “lifespan”.

All in all, blockchain-based generative art reimagines what it means to own, view, and live with art. It allows artists and programmers to create living artworks that lives beyond the lightfastness of the paint, or the durability of the material. The art continuously lives on instead of deteriorating, and invites new interaction over time. This shifts the emphasis of “the final result” to the process or in this case, the system.

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[4] Cam Thompson, Eli Tan, “Generative-Art NFTs Bring New Heat to Crypto Winter,” *CoinDesk*, October 29, 2022, https://www.coindesk.com/web3/2022/10/29/generative-art-nfts-bring-new-heat-to-crypto-winter.

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